Time-Varying Nonlinear Factor Models via Deep Autoencoders

Preliminary draft

Matthew Ladd

October 20, 2026

1 Introduction

The relationship between household consumption, disposable income, and credit constraints is not fixed. When credit is easily available, an additional dollar of disposable income might raise consumption by 80 cents; when credit is tight, the response may be only 30. Within either setting the response is not linear: a single dollar is used differently depending on whether the household is already liquidity-constrained. Zooming out to thousands of goods across thousands of households, the consumption of each good in each household likely responds to a small number of shared factors (income, credit conditions, prices). Exposure to those factors could reasonably be nonlinear and changing over time.

Factor models summarize this idea. A cross section of N observed variables is driven by a small number of latent factors $F_t \in \mathbb{R}^K$, with each unit's exposure specified as a map from the factors to the observation. The earliest and still dominant specification assumes the map is linear and fixed over the sample: $X_{it} = \lambda_i' F_t + U_{it}$. Linearity is a convenience rather than a theoretical requirement, and constant loadings are untenable over a long horizon. When either assumption fails, the recovered common component is biased, and any downstream analysis inherits the bias.

Two recent literatures relax these assumptions separately. Shen and Xiu (2026) allow the loading to be an unknown nonlinear function of the factors and propose a deep autoencoder estimator, with non-asymptotic guarantees and rates that match the minimax bound for the nonparametric problem. Su and Wang (2017) allow the loading to vary smoothly over time and propose a local principal component estimator, with asymptotic theory for the estimated factors, loadings, and common components. Neither handles both simultaneously. When loadings are nonlinear in the factors and also shift with the regime, relaxing one assumption and leaving the other in place does not eliminate the bias.

The gap. The two extensions are logically independent, and each has been studied in isolation. The following table maps the literature: The bottom-right cell is the object of this paper.

	Linear loadings	Nonlinear loadings
Static	Stock and Watson (2002); Bai (2003)	Shen and Xiu (2026)
Time-varying	Su and Wang (2017)	This paper

This paper. The paper has three contributions.

Theory. A non-asymptotic rate of convergence for the mean squared error of the recovered common component, in the case where loadings are nonlinear and time-varying. The rate is $T^{-2\beta/(2\beta+K+1)} + N^{-1}K_1 + \epsilon_N$, where β is the smoothness of the loading function, K is the factor dimension, N and T are the cross-sectional and time-series sizes, $K_1 \geq K$ is the bottleneck width, and ϵ_N is the encoder approximation error. The rate nests the static-linear rate of Bai (2003) and the static-nonlinear rate of Shen and Xiu (2026) as special cases, and strictly dominates the rate of a local autoencoder analogue of Su and Wang (2017).

Method. A global autoencoder whose encoder and decoder both take the scaled time index $s_t = t/T$ as an input. The global parametrization shares weights across time, which is what makes the model trainable at realistic sample sizes. A version that fits a separate autoencoder at each time period does not train at $T = 250$.

Evidence. Simulations and four financial panels. The estimator wins where the loading function is nonlinear and time-varying, and ties where it is linear.

Preview of findings. On simulated data with nonlinear, time-varying loadings, the estimator reduces mean squared error of the recovered common component by 20–25% relative to the best static method, with the gap narrowing as T grows but remaining positive at every sample size considered. On linear, static data, there is no gain, as the theory predicts. On a panel with apparently nonlinear loadings (crypto returns), out-of-sample R^2 improves by 3.6 percentage points over static PCA, significant at the 5% level under a paired bootstrap over refit windows. On panels with apparently linear loadings (equity returns, equity volatility, crypto volatility), R^2 is unchanged.

The 3.6-point gain is statistically significant but has not been shown to translate into economic value. Whether it improves forecasts, pricing, or portfolio decisions is the focus of future work, and the paper returns to this open question in Section 7.

Related literature. The paper sits at the intersection of four literatures.

Linear factor models. The modern treatment of large-dimensional factor models originates with Chamberlain and Rothschild (1983) and Connor and Korajczyk (1986). The asymptotic theory for principal component estimation is developed by Stock and Watson (2002), Bai and Ng (2002), and Bai (2003). This literature assumes loadings that are linear in the factors and constant over time.

Time-varying factor models. Stock and Watson (2009) show that small instabilities in loadings leave PCA estimates consistent, but larger instabilities can lead to substantial misestimation. Breitung and Eickmeier (2011) and Chen, Dolado, and Gonzalo (2014) propose tests for one-time structural breaks. Su and Wang (2017) introduce the local principal component estimator, with loadings modelled as smooth functions of the scaled time index. The

local estimator pays a kernel-smoothing penalty: the nonparametric rate for the loading is $(Th)^{-2\beta/(2\beta+K+1)}$ at each reference time, and the effective sample per fit is only Th. The present paper avoids this penalty by parametrizing time through a neural-network input rather than kernel weighting. The rate improves from $T^{-2\beta/(2\beta+K+2)}$ to $T^{-2\beta/(2\beta+K+1)}$.

Nonlinear factor models and autoencoders. Gu, Kelly, and Xiu (2021) introduce a conditional autoencoder that allows factor exposures to depend on observable characteristics. Shen and Xiu (2026) provide the non-asymptotic theoretical foundation for autoencoder factor models with constant loadings. This paper extends Shen and Xiu (2026) by adding time variation in the loading function.

Deep learning theory. Schmidt-Hieber (2020), Kohler and Langer (2021), Farrell, Liang, and Misra (2021), and Jiao et al. (2023) provide the approximation and concentration toolbox used in the proofs.

2 Related Literature

The preceding paragraphs summarize the four streams. This section briefly notes two additional connections. Gu, Kelly, and Xiu (2021) apply the conditional autoencoder to US equity returns and find substantial gains over linear factor models. The present paper's characteristic-conditioned extension (Appendix B) nests that framework as a special case, though the empirical work here does not exercise it. Stock and Watson (2009) document that loadings in US macro data exhibit structural instability; Su and Wang (2017) test this formally and reject constancy. The present paper takes those findings as motivation for the time-varying specification and does not re-test the constancy of loadings.

3 The Model

3.1 A nonlinear, time-varying factor model

Let $\{X_{it}\}$ denote a panel indexed by $i = 1, \ldots, N$ units and $t = 1, \ldots, T$ periods. Following the nonlinear factor framework introduced by Amemiya and Yalcin (2001), extended to accommodate time-varying factor loadings, the observed data are assumed to admit the decomposition

$$X_{it} \;=\; X_{it}^{\star} + U_{it} \;=\; \varphi_i^{\star}(F_t^{\star},\, t/T) + U_{it}, \qquad i = 1, \ldots, N, \quad t = 1, \ldots, T, \tag{1}$$

where $X_{it}^{\star}$ denotes the common component, $\varphi_i^{\star}(\cdot,\cdot) : \mathbb{R}^K \times [0,1] \to \mathbb{R}$ is the loading function of unit i, $F_t^{\star} \in \mathbb{R}^K$ is a K-dimensional vector of latent factors, and U_{it} is an idiosyncratic error term.

Two departures from the standard linear static model are important. First, $\varphi_i^{\star}(\cdot, s)$ is an unknown function rather than a linear map from the factors to the observation. Second, the loading function depends on $s = t/T$.

Model (1) nests three specifications that have been studied in isolation: the static linear model ($\varphi_i^{\star}(F, s) = \Lambda_i' F$), the time-varying linear model of Su and Wang (2017) ($\varphi_i^{\star}(F, s) = \lambda_i(t/T)'F$), and the static nonlinear model of Shen and Xiu (2026) ($\varphi_i^{\star}(F, s) = \varphi_i(F)$).

Why rescale time to $[0,1]$, and why smoothness. Setting $s = t/T \in [0,1]$ places time on a fixed support. For a fixed s, the sample in a neighborhood of $t = sT$ grows with T, which is what makes the loading identifiable in the limit (Su and Wang, 2017). The loading φ_i is assumed β-smooth in both (F, s): its derivatives up to order β exist and are bounded. This is the smoothness class targeted by neural-network approximation theory (**stone1982optimal**; Kohler and Langer, 2021).

The estimand. The common component $X^\star$. As in every factor model, the factors F_t and loadings φ_i are not separately identifiable; only the common component they produce is. That is what is targeted throughout.

3.2 Assumptions

Assumption 1 (Boundedness and noise)**.** *The factors $\{F_t^\star\}_{t=1}^T$ are supported on the compact box $\mathcal{B} = [-B, B]^K$ with $B \geq 1$ almost surely. Conditional on $F^\star$, $\mathrm{vec}(U) \overset{d}{=} \Sigma^{1/2}\mathrm{vec}(Z)$, where $Z \in \mathbb{R}^{N\times T}$ has independent, zero-mean, sub-Gaussian entries with sub-Gaussian norm bounded by σ_z, and Σ is positive semidefinite with bounded spectral norm.*

Assumption 2 (Smoothness)**.** *There exists $\beta \in \mathbb{N}_+$ with $\beta \geq 1$ such that $\varphi_i^\star \in \mathcal{H}^\beta(\mathcal{B} \times [0,1], B)$ for each i.*

Assumption 3 (Encoder approximation class)**.** *There exist a sparse linear map $W^\star \in \mathbb{R}^{K\times N}$ with $\|W^\star\|_0 \leq L_N K$ and $\|W^\star\varphi^\star\|_\infty \leq B$, and a map $\eta^\star : [-2B, 2B]^K \times [0,1] \to \mathbb{R}^K$ with $\eta^\star \in (\mathcal{H}^\beta([-2B, 2B]^K \times [0,1], C_\eta))^K$ for a finite constant $C_\eta = C_\eta(\beta, K, B)$ that is uniform in N, such that*

$$\sup_{(x,s)\in\mathcal{B}\times[0,1]} \|\eta^\star(W^\star\varphi^\star(x,s), s) - x\|^2 + \|W^\star\|^2 \;\leq\; 2\epsilon_N. \tag{2}$$

Assumption 4 (Weak dependence)**.** *The process $\{(F_t^\star, U_t)\}_{t=1}^\infty$ is strictly stationary, β-mixing with $\beta(u) \leq e^{-\kappa u}$ for some $\kappa > 0$, and ρ-mixing with $\sum_{u=1}^\infty \rho(u) < \infty$.*

The theory-practice gap (sparsity implemented via weight decay, optimization error) is discussed in Appendix D.

3.3 An autoencoder is a nonlinear factor model

An autoencoder is a neural network trained to reconstruct its own input through a low-dimensional bottleneck layer. The network has two components: an encoder ρ that maps the input $X_t \in \mathbb{R}^N$ to a bottleneck vector in $\mathbb{R}^{K_1}$, and a decoder that maps the bottleneck back to the observation space.

When the encoder and decoder are linear and the loss is squared error, the autoencoder is equivalent to principal component analysis (Baldi and Hornik, 1989). The bottleneck width K_1 corresponds to the number of principal components. With nonlinear activations, the same architecture becomes a nonlinear factor model: the bottleneck plays the role of F_t, and the decoder plays the role of φ_i.

Two distinctions from PCA are important. First, the number of factors K is not identified in nonlinear factor models, whereas in linear factor models it is identified under standard pervasiveness conditions (**"S 3.2**; Shen and Xiu, 2026). In practice, the bottleneck width $K_1 \geq K$ is treated as a hyperparameter, chosen by validation. Over-specification adds a linear $N^{-1}K_1$ penalty to the rate; it does not break the estimator. Second, the autoencoder is unsupervised: no external target is required. The common component $X^\star$ is the object of interest, and the encoder learns a low-dimensional representation of it directly from the input data.

3.4 The global time-conditioned autoencoder

The primary estimator of this paper is the global time-conditioned autoencoder (TV-AE-G). It is a single autoencoder in which the scaled time index $s = t/T$ enters as an input to both the encoder and the decoder. The encoder and decoder classes are

$$\rho \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, C, B), \qquad \psi_i \in \mathcal{F}_{K_1+1}^{1}(d_2, w_2, C, B), \tag{3}$$

where $\mathcal{F}_{n_0}^{n_{d+1}}(d, w, C, B)$ is the ReLU network class of Shen and Xiu (2026), §2.2. The model is

$$X_{it} \;=\; \psi_i(\rho(X_t, t/T),\, t/T) + U_{it}. \tag{4}$$

The estimator minimizes the unweighted loss

$$(\hat{\rho}, \{\hat{\psi}_i\}_{i=1}^N) = \arg\min \sum_{t=1}^{T} \sum_{i=1}^{N} \left(X_{it} - \psi_i(\rho(X_t, t/T), t/T)\right)^2, \tag{5}$$

over the disjoint-output class $\mathcal{F}_{AE}^{K_1}$ of Shen and Xiu (2026), extended to $(N+1)$- and (K_1+1)-dimensional inputs. The estimated common component is $\hat{X}_{it}^G := \hat{\psi}_i(\hat{\rho}(X_t, t/T), t/T)$.

Per-output decoders. The decoder consists of N separate subnetworks, one per output, rather than a single fully-connected decoder. Shen and Xiu (2026) show that this preserves the approximation rate while reducing the parameter count by a factor of N. This is what keeps the model trainable at small T.

Time as input. The time index $s_t = t/T$ enters as an input to both the encoder and the decoder. The decoder captures time-varying loadings. The encoder captures time-varying factor composition: which units dominate the cross-section shifts over time, and the encoder's input must reflect that. A single model is trained on all T periods, sharing parameters across time.

Why global rather than local. A local autoencoder analogue of Su and Wang (2017) would fit a separate autoencoder at each grid point in t/T, weighted by a kernel. At $T = 250$, the effective sample per grid point is roughly $Th \approx 30$ observations, far below what a network with thousands of parameters needs to train. The global parametrization shares parameters across time, and the effective sample is T rather than Th. This is what makes the estimator usable at realistic T. The local analogue's theoretical properties are characterized in Appendix A, and its finite-sample failure is documented in Appendix F.

3.5 Identification

The factors and loading functions in (1) are not separately identified. For any time-indexed family of invertible transformations $\mu_t : \mathbb{R}^K \to \mathbb{R}^K$, the reparameterization

$$F_t \mapsto \mu_t(F_t), \qquad \varphi_i^\star(\cdot, t/T) \mapsto \varphi_i^\star(\mu_t^{-1}(\cdot), t/T) \tag{6}$$

leaves the common component $X_{it}^\star$ unchanged. This generalizes the rotation indeterminacy of static linear factor models: μ_t may be a nonlinear and time-varying map. The common component $X_{it}^\star$ is identified, and all convergence results in Section 4 are stated directly for $X_{it}^\star$. The factor series $\{\hat{F}_t\}$ is identified only up to a time-varying invertible transformation; comparisons of $\hat{F}_t$ to $F_t^\star$ therefore require a nonlinear alignment. The common-component mean-squared error is used as the primary metric throughout, precisely because it sidesteps this ambiguity. A further result on factor recovery is in Appendix C.

4 Theory

This section states the main theoretical result. The proof is in Appendix D. Throughout, $a_{N,T} \lesssim b_{N,T}$ means $a_{N,T} \le C b_{N,T}$ for a constant C independent of N and T, and $s_t = t/T$, $X_{it}^\star = \varphi_i^\star(F_t^\star, s_t)$.

4.1 The main theorem

Theorem 1 (In-sample recovery)**.** *Suppose Assumptions 1–3 hold and* $\log T = o(\epsilon_N^{-1})$*. Let* $K \le K_1 \le \min(w_1, w_2)$*,* $d_1 \asymp d_2 \asymp \log T$*, and* $w_1 \asymp w_2 \asymp T^{(K+1)/(2(2\beta+K+1))}$*. Then for sufficiently large* $\min(N, T)$*, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{NT}\sum_{t=1}^{T}\sum_{i=1}^{N}\left(\hat{X}_{it}^{\mathrm{G}} - X_{it}^\star\right)^2 \;\lesssim\; \left(T^{-\frac{2\beta}{2\beta+K+1}} \;+\; N^{-1}K_1 \;+\; \epsilon_N\right)\log^3 T. \tag{7}$$

Reading the rate. The rate has three terms. The first, $T^{-2\beta/(2\beta+K+1)}$, is the nonparametric rate for estimating the loading function $\varphi_i^\star(F, s)$ in $K+1$ effective dimensions. It is the smallest error any method could achieve, even if the factors were observed. It is smaller when β is large (smoother loadings) and K is small (fewer dimensions). The second, $N^{-1}K_1$, is the parametric error from estimating K_1 factor values at each of T dates. It shrinks with N (more cross-sectional data per factor) and grows with K_1 (more factors to estimate). The third, ϵ_N, is the encoder approximation error from Assumption 3. It is small when the factors are strong and the loadings are close to linear.

The rate tells you how quickly accuracy improves as N and T grow, and which of the two dimensions is more valuable for a given problem. It also says that the recovery error is a lower bound on the error of any downstream task that uses the recovered common component.

4.2 Comparison to prior work

The rate compares to prior work as follows. †At $\beta = 2$, $K = 2$. Larger magnitude means

Method	Rate	Exponent†
Shen and Xiu (2026) (static)	$T^{-2\beta/(2\beta+K)}$	-0.667
This paper	$T^{-2\beta/(2\beta+K+1)}$	-0.571
Local analogue of Su and Wang (2017)	$T^{-2\beta/(2\beta+K+2)}$	-0.500

faster decay as T grows.

The rate is slower than Shen and Xiu (2026) because the loading function lives in one more dimension (t/T). It is faster than the local analogue of Su and Wang (2017) because there is no kernel-smoothing penalty. The local analogue's rate is derived in Appendix A. In the linear case with $\varphi_i^\star(F, s) = \lambda_i(s)'F$ and $L_N \asymp N$, the rate reduces to $N^{-1}K_1 + T^{-2\beta/(2\beta+K+1)}$, matching the classical rate up to the extra time dimension.

5 Simulations

5.1 Design

Each data-generating process tests one cell of the 2×2 contribution grid, plus a boundary case. The linear/constant DGP is a placebo: PCA is the correct model, and the autoencoder

	Linear loadings	Nonlinear loadings
Static	linear/constant	exponential/constant
Time-varying	linear/logistic	exponential/logistic

should match it. The linear/logistic DGP has local PCA as the correctly specified model. The exponential/constant DGP has the static autoencoder of Shen and Xiu (2026) as the correctly specified model. The exponential/logistic DGP is the main test: neither linear model fits. The quadratic/logistic DGP is a boundary case, with a mostly static nonlinearity and a small time-varying component.

Setup: $N = 100$; $K = 2$ (matching the Fama-French three-factor model after subtracting the market); $T \in \{200, 500, 1000, 2000\}$. The latent factors $F_t^\star \in \mathbb{R}^K$ are drawn independently across t from the uniform distribution on $[-2, 2]^K$. Each unit's common component is normalized to unit variance; the idiosyncratic noise is i.i.d. standard normal in the main-text runs. The time-varying case uses $\lambda_i(t/T) = \lambda_{i0} + b\, g(t/T)$, with g the logistic transition $g(s) = 1/(1 + \exp(-\kappa(s - 1/2)))$, $\kappa = 10$, $b = 1$. Replication counts: 10 at $T = 200$; 5 at $T = 500$ and $T = 1000$; 3 at $T = 2000$ for the quadratic DGP.

Four estimators are compared: static PCA, the local PCA estimator of Su and Wang (2017), the static autoencoder of Shen and Xiu (2026), and the global time-conditioned autoencoder of this paper. All autoencoders use the disjoint-output architecture and are trained by Adam with bottleneck $K_1 = 3$, encoder width 64, decoder width 32, learning rate

10^{-3}, weight decay 10^{-3}, 500 epochs max, early-stopping patience 30, and ensemble size 3. The hyperparameters were selected by a capacity sweep reported in Appendix H.

5.2 Results

Table 1 reports the mean MSE across replications.

Table 1: Mean squared error of the recovered common component. Table entries are mean MSE across replications; lower is better.

DGP	T	PCA	Local PCA	Static AE	Time-conditioned AE
Linear, constant	200	0.034	0.067	0.048	0.048
	1000	0.024	0.033	0.028	0.028
Exponential, logistic	200	0.192	0.178	0.112	**0.088**
	500	0.183	0.154	0.072	**0.059**
	1000	0.202	0.160	0.050	**0.046**
Quadratic, logistic	200	0.460	0.442	0.160	**0.146**
	500	0.431	0.439	0.105	**0.093**
	1000	0.448	0.461	0.086	0.087
	2000	0.459	0.442	**0.070**	0.079

Placebo (linear/constant). The autoencoder does not gain where there is nothing to gain. PCA wins at small T (fewer parameters); the autoencoders catch up as T grows. The two autoencoders tie: the time input costs nothing when unused.

Main test (exponential/logistic). The autoencoder wins by 20–25% at $T = 200$ and $T = 500$ relative to the best static method, with the gap narrowing to 8% at $T = 1000$ but remaining positive. PCA plateaus at 0.19–0.20 across all sample sizes, reflecting its irreducible misspecification bias. Both autoencoders improve with T; the time-conditioned autoencoder stays ahead throughout.

Boundary case (quadratic/logistic). At $T \leq 500$, the time-conditioned autoencoder wins. At $T = 1000$ the two autoencoders tie. At $T = 2000$ the static autoencoder overtakes. The dominant nonlinearity is static; the static autoencoder learns it very accurately once T is large. The time-varying component contributes little marginal value at large T, and the time-conditioned autoencoder continues to pay a variance cost for estimating the time input. At what point the cost dominates the benefit depends on the DGP; other explanations (estimation noise, sample-specific features) are also possible. This is a genuine bias-variance tradeoff, not a failure of the estimator.

Additional runs. The full DGP grid, the first-pass baseline on seven DGPs, and the symmetric-regularization correction are documented in Appendix H.

6 Empirical Work

6.1 Four-panel design

The empirical work evaluates the theory on real panels. The theory (Theorem 1) constrains the accuracy with which the estimator recovers the common component $X^\star$. The panels are chosen so that the theory makes a prediction for each: if the estimator captures genuine nonlinearity, it should win on the nonlinear panel and tie on the linear panels.

Panel	Expect nonlinear?	Why
Equity returns	No	Linear factor structure is well documented
Equity volatility	No	Dominated by a single volatility factor
Crypto returns	**Yes**	Extreme returns, regime changes, large differences across coins
Crypto volatility	No	Driven by a small number of linear factors

Out-of-sample R^2. This measures how well the recovered common component predicts held-out data. 1 is perfect; 0 is no better than predicting zero; negative is worse than that. The full definition and its relation to the theoretical quantity are in Appendix I.

Setup. Rolling window $T_{\text{est}} = 250$ days, refit every 21 days. Bottleneck $K_1 = 5$ factors, matching the PCA benchmark. Out-of-sample window: 2010–2024 (equity), 2021–2024 (crypto; the panel is too thin before 2021). Data construction, K_1 selection, and the unbalanced-panel treatment are in Appendix I.

6.2 Results

Table 2: Pooled out-of-sample R^2, four panels. $N = 100$ stocks or $N = 50$ coins; $T_{\text{est}} = 250$; $H = 21$.

Panel	PCA	Static AE	AE with time
Equity returns	0.578	0.574	0.573
Equity volatility	0.989	0.989	0.985
Crypto returns	0.606	0.626	**0.642**
Crypto volatility	0.972	0.972	0.966

On the equity panels, the autoencoder is statistically indistinguishable from PCA (the differences are less than one percentage point). On the crypto volatility panel, the same holds:

Table 3: Paired bootstrap over refit windows on the crypto return panel (2,000 draws).

Comparison	ΔR^2	95% CI
AE with time − PCA	+0.036	[+0.008, +0.075]
AE with time − Static AE	+0.015	[−0.001, +0.045]
Static AE − PCA	+0.021	[+0.007, +0.036]

both methods achieve R^2 above 0.97. On the crypto return panel, the ordering changes: the static autoencoder gains 2.0 percentage points over PCA, and the time-conditioned autoencoder gains another 1.6 points, for a total of 3.6 points over PCA.

Table 3 reports the paired bootstrap. The nonlinear autoencoder beats PCA by 3.6 percentage points (significant at 5%). The time input adds 1.5 points over the static autoencoder (directionally positive; the CI marginally includes zero). All three autoencoders tie PCA on the three linear-structure panels.

6.3 Robustness

Per-window distribution. The gain could be driven by a few outlier windows; it is not. Both the mean and median R^2 are higher than PCA (0.62 vs. 0.60; 0.63 vs. 0.61). The standard deviation across windows is lower (0.16 vs. 0.18), so the estimator is also more stable.

Year by year. The gain appears in every year of the out-of-sample period, but is concentrated in 2021. That year's +11.8pp is consistent with the theory: the time input should matter most when the loading structure shifts, and 2021 was the largest crypto regime change in the sample. In 2022–2024, gains range from 0.4 to 3.5pp.

Bottleneck width. If the estimator were only overfitting noise, the gain would rise with K_1. Instead, it peaks at $K_1 = 2$ (a gain of 3.4pp) and falls to zero at $K_1 = 12$. That pattern is what genuine low-dimensional nonlinear structure would produce.

Additional tests. A one-sided test for the presence of nonlinearity, applied across all four panels, rejects linearity only on crypto returns; the equity return, equity volatility, and crypto volatility panels do not reject. Details, and the known limitation that the test cannot distinguish time-varying linear from nonlinear alternatives, are in Appendix G.

7 Discussion

The 3.6-point gain is a statistical improvement. It has not been shown to translate into economic value. The most direct test of downstream economic value is the minimum-variance portfolio application, which comes back negative: the autoencoder does not improve MVP Sharpe. That result is reported in Appendix E and is informative about the boundary of what the estimator can deliver.

What the paper establishes	What remains open
A rate for the MSE of the recovered common component when loadings are nonlinear and time-varying.	**Economic significance.** The +3.6pp has not been mapped to a decision-relevant improvement. Whether it translates to better forecasts, pricing, or portfolio decisions is the open question.
The rate is faster than the natural local alternative.	**No lower bound.** The rate is an upper bound; whether any estimator can do better is unknown.
Simulations: gains where nonlinearity and time variation are both present; ties where they are not.	**No inference.** No confidence bands on $X^\star$, no tests for loadings.
Empirics: +3.6pp out-of-sample R^2 on crypto returns (significant at 5%); no loss on three linear-structure panels.	**Thin crypto panel.** 50 coins, roughly 70 refit windows.

8 Extensions

Three next steps are natural.

A forecasting exercise. The most direct test of economic value is to use the recovered common component as a predictor. The current empirical design freezes the time index at its last training value during the test window, which is a simplification; a proper time-extrapolation scheme at the test boundary is a prerequisite for a genuine out-of-sample forecasting exercise. The reconstructed common component is the natural input to a factor-augmented forecasting regression, and the reconstruction-accuracy gain documented here should translate into forecast-accuracy gains if the loading structure is genuinely nonlinear and time-varying.

Longer or thicker panels. The crypto panel is thin: 50 coins from 2021 onwards, roughly 70 refit windows. A longer panel would sharpen the comparison and allow a proper test of the time input's contribution over the static autoencoder. Other panels with suspected nonlinear loadings (option-implied volatility, cross-country equity indices, commodity futures) would provide additional test settings.

Characteristic-conditioned empirical work. Gu, Kelly, and Xiu (2021) show that conditioning asset-pricing models on observable characteristics improves performance. A characteristic-conditioned version of the present estimator, defined in Appendix B, nests that framework and adds time variation to the loading function. Testing whether the conditional extension inherits the Gu, Kelly, and Xiu (2021) gains, and whether it adds further gains from time variation, is a natural next step. The price-based characteristics available in the

current data span less than 2% of return variance and cannot support this test; the OSAP or WRDS characteristic panel would.

9 Conclusion

The question, the gap, and the contribution. The question is how much allowing factor loadings to be nonlinear and time-varying improves the recovery of common components. Shen and Xiu (2026) handle nonlinearity; Su and Wang (2017) handle time variation; neither handles both. When both are present, both leave the recovered common component biased. The paper proposes a global autoencoder that takes $s_t = t/T$ as an input to both the encoder and the decoder. The estimator is theoretically tractable, trainable at realistic sample sizes, and consistent under the joint nonlinear, time-varying model.

Findings and next steps. The paper establishes a rate for the MSE of the recovered common component, faster than the natural local alternative. In simulations, the estimator delivers a 20–25% reduction in MSE where nonlinearity and time variation are both present (narrowing to 8% at $T = 1000$), and ties elsewhere. Empirically, it delivers a +3.6pp out-of-sample R^2 improvement on crypto returns (significant at 5%) and no loss on three linear-structure panels.

The next step is a forecasting exercise that uses the recovered common component as a predictor, with a proper time-extrapolation scheme at the test boundary. This is the natural setting where reconstruction accuracy should translate into economic value.

References

Amemiya, Yasuo and Ilker Yalcin (2001). "Nonlinear Factor Analysis as a Statistical Method". In: *Statistical Science* 16.3, pp. 275–294.

Bai, Jushan (2003). "Inferential Theory for Factor Models of Large Dimensions". In: *Econometrica* 71.1, pp. 135–171.

Bai, Jushan and Serena Ng (2002). "Determining the Number of Factors in Approximate Factor Models". In: *Econometrica* 70.1, pp. 191–221.

Baldi, Pierre and Kurt Hornik (1989). "Neural Networks and Principal Component Analysis: Learning from Examples Without Local Minima". In: *Neural Networks* 2.1, pp. 53–58.

Breitung, Jörg and Sandra Eickmeier (2011). "Testing for Structural Breaks in Dynamic Factor Models". In: *Journal of Econometrics* 163.1, pp. 71–84.

Chamberlain, Gary and Michael Rothschild (1983). "Arbitrage, Factor Structure, and Mean-Variance Analysis on Large Asset Markets". In: *Econometrica* 51.5, pp. 1281–1304.

Chen, Liang, Juan J. Dolado, and Jesús Gonzalo (2014). "Detecting Big Structural Breaks in Large Factor Models". In: *Journal of Econometrics* 180.1, pp. 30–48.

Connor, Gregory and Robert A. Korajczyk (1986). "Performance Measurement with the Arbitrage Pricing Theory: A New Framework for Analysis". In: *Journal of Financial Economics* 15.3, pp. 373–394.

Farrell, Max H., Tengyuan Liang, and Sanjog Misra (2021). "Deep Neural Networks for Estimation and Inference". In: *Econometrica* 89.1, pp. 181–213.

Gu, Shihao, Bryan Kelly, and Dacheng Xiu (2021). "Autoencoder Asset Pricing Models". In: *Journal of Econometrics* 222.1, pp. 429–450.

Jiao, Yuling et al. (2023). "Deep Nonparametric Regression on Approximate Manifolds: Nonasymptotic Error Bounds with Polynomial Prefactors". In: *Annals of Statistics* 51.2, pp. 691–716.

Kohler, Michael and Sophie Langer (2021). "On the Rate of Convergence of Fully Connected Deep Neural Network Regression Estimates". In: *Annals of Statistics* 49.4, pp. 2231–2249.

Schmidt-Hieber, Johannes (2020). "Nonparametric Regression Using Deep Neural Networks with ReLU Activation Function". In: *Annals of Statistics* 48.4, pp. 1875–1897.

Shen, Zhouyu and Dacheng Xiu (2026). *Deep Autoencoders for Nonlinear Factor Models: Theory and Applications*. Chicago Booth Research Paper 25-14. Working paper; original draft 2024. University of Chicago, Booth School of Business.

Stock, James H. and Mark W. Watson (2002). "Forecasting Using Principal Components from a Large Number of Predictors". In: *Journal of the American Statistical Association* 97.460, pp. 1167–1179.

— (2009). "Forecasting in Dynamic Factor Models Subject to Structural Instability". In: *The Methodology and Practice of Econometrics: A Festschrift in Honour of David F. Hendry*. Ed. by Jennifer Castle and Neil Shephard. Oxford University Press, pp. 173–205.

Su, Liangjun and Xia Wang (2017). "On Time-Varying Factor Models: Estimation and Testing". In: *Journal of Econometrics* 198.1, pp. 84–101.

A The local autoencoder

This appendix documents the local autoencoder (TV-AE-L), the direct nonlinear analogue of Su and Wang (2017). It is included for comparison and to characterize the kernel-smoothing penalty that motivates the global parametrization in the main text.

A.1 Architecture and loss

Let $\{x_1, \ldots, x_m\} \subset (0,1)$ be a grid of reference points with uniform spacing $O(h)$, and let K_h be the Epanechnikov kernel with bandwidth h. At each grid point x_j, the local autoencoder minimizes

$$(\hat{\rho}_j, \{\hat{\psi}_{i,j}\}_{i=1}^N) = \arg\min \sum_{t=1}^{T} K_h\left(\frac{t/T - x_j}{h}\right) \sum_{i=1}^{N} \left(X_{it} - \psi_i(\rho(X_t), x_j)\right)^2 . \tag{8}$$

The reconstruction at time t averages across grid points with the same kernel weights.

A.2 Rate

Proposition 1 (In-sample rate for TV-AE-L)**.** *Suppose the conditions of Theorem 1 hold and let* $h \asymp T^{-1/(2\beta+K+2)}$*. Then, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{NT}\sum_{t,i}\big(\hat{X}_{it}^{L} - X_{it}^{\star}\big)^2 \;\lesssim\; \big(T^{-\frac{2\beta}{2\beta+K+2}} + N^{-1}K_1 + \epsilon_N\big)\log^4 T, \tag{9}$$

where $\hat{X}_{it}^{L}$ *is the local autoencoder reconstruction.*

The rate $T^{-2\beta/(2\beta+K+2)}$ is strictly slower than the TV-AE-G rate $T^{-2\beta/(2\beta+K+1)}$ because the exponent in the denominator is $2\beta + K + 2$ rather than $2\beta + K + 1$. The gap is the kernel-smoothing penalty of local estimation, and it is the technical reason the empirical work of Section 6 uses TV-AE-G.

A.3 Finite-sample behavior at small T

A local autoencoder analogue of Su and Wang (2017) was implemented and evaluated on the same simulation grid. At $T = 200$ and $N = 100$, with the Silverman bandwidth $h \approx 0.148$, the effective sample at each grid point is $Th \approx 30$ observations. A network with roughly 8,000 parameters trained on 30 observations does not learn. In the first run (10 grid points, 2 seeds, 300 epochs), the estimator produced an MSE of 0.1000 on the linear/constant DGP (vs. 0.0483 for the static autoencoder) and 0.1885 on the quadratic/logistic DGP (vs. 0.1701 for the static autoencoder).

Diagnostics confirmed the cause. Effective training sample per grid was 48 observations in the interior and 7 at the boundaries. Early stopping fired at 60–136 epochs. The model was not overtrained; it was undertrained due to data scarcity. This is a fundamental property of local nonparametric estimation at $T = 200$, not a tuning issue, and it is consistent with the rate comparison in Proposition 1: the local estimator pays a kernel-smoothing penalty, and at $T = 200$ the penalty dominates. This finding is the practical motivation for the global estimator.

B The characteristic-conditioned autoencoder

This appendix documents the characteristic-conditioned autoencoder (TV-AE-C), the extension of TV-AE-G to observable characteristics in the spirit of Gu, Kelly, and Xiu (2021). It is included for completeness; the empirical work of Section 6 does not exercise it.

B.1 Architecture

Let $Z_{i,t-1} \in \mathbb{R}^P$ denote unit i's characteristics measured at $t-1$. The model is

$$r_{it} \;=\; \psi_i(\rho(X_t, t/T),\, Z_{i,t-1},\, t/T) + u_{it}, \tag{10}$$

where $\rho : \mathbb{R}^N \times [0,1] \to \mathbb{R}^{K_1}$ is the encoder and each $\psi_i : \mathbb{R}^{K_1} \times \mathbb{R}^P \times [0,1] \to \mathbb{R}$ is a decoder. The target loading function is $\varphi_i^{\star} : \mathcal{B} \times [-B_Z, B_Z]^P \times [0,1] \to \mathbb{R}$, assumed β-Hölder in its arguments, with $\|Z_{i,t-1}\|_\infty \le B_Z$ almost surely.

Model (10) nests Gu, Kelly, and Xiu (2021) as the special case with no t/T input, TV-AE-G as the special case with $P = 0$, and a static linear factor model when both inputs are dropped and the decoders are linear.

B.2 Rate

Proposition 2 (In-sample rate for TV-AE-C)**.** *Suppose the conditions of Theorem 1 hold. Suppose in addition that* $\|Z_{i,t-1}\|_\infty \le B_Z$ *almost surely and that the target loading function* $\varphi_i^\star : \mathcal{B} \times [-B_Z, B_Z]^P \times [0,1] \to \mathbb{R}$ *lies in* $\mathcal{H}^\beta(\mathcal{B} \times [-B_Z, B_Z]^P \times [0,1], B)$ *for each* i*. Let the decoder class be* $\mathcal{F}^1_{K_1+P+1}(d_2, w_2, C, B)$*, with* $d_2 \asymp \log T$ *and* $w_2 \asymp T^{(K+P+1)/(2(2\beta+K+P+1))}$*. Then for sufficiently large* $\min(N,T)$*, with probability at least* $1-C\exp(-cT)-C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{NT}\sum_{t,i}\big(\hat{X}_{it}^{\mathrm{C}} - X_{it}^\star\big)^2 \;\lesssim\; \big(T^{-\frac{2\beta}{2\beta+K+P+1}} + N^{-1}K_1 + \epsilon_N\big)\log^3 T, \tag{11}$$

where $\hat{X}_{it}^{C} := \hat{\psi}_i(\hat{\rho}(X_t, s_t), Z_{i,t-1}, s_t)$*.*

B.3 Cost of conditioning

Proposition 2 recovers Theorem 1 when $P = 0$. The P-dimensional cost is $T^{-2\beta/(2\beta+K+P+1)}$ instead of $T^{-2\beta/(2\beta+K+1)}$. For $P = 6$, $\beta = 2$, $K = 2$, this is $T^{-0.36}$ vs. $T^{-0.57}$: a real slowdown. The tradeoff is that the characteristics may absorb part of the loading-function variation that would otherwise be attributed to the nonlinear factor structure. If some components of $Z_{i,t-1}$ are discrete or lie on a lower-dimensional subset, the effective dimension is smaller than P and the rate is correspondingly faster.

C Factor recovery

Proposition 3 (Factor recovery for TV-AE-G)**.** *Under the conditions of Theorem 1, there exists, for each* $t = 1, \dots, T$*, a map* $\mu_t : \mathbb{R}^{K_1} \to \mathbb{R}^K$ *(depending on the trained encoder, the architecture, and the DGP, but not on data realizations beyond the encoder) such that, with probability at least* $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$*,*

$$\frac{1}{T}\sum_{t=1}^{T}\|\mu_t(\hat{\rho}(X_t, s_t)) - F_t^\star\|^2 \;\lesssim\; \big(\epsilon_N K_1 + N\epsilon_N T^{-\frac{2\beta}{2\beta+K+1}} + N\epsilon_N^2\big)\log^4 T. \tag{12}$$

Remark 1. *The map* μ_t *is not the identity in general. The estimated factors* $\hat{\rho}(X_t, s_t)$ *are identified only up to a nonlinear, time-indexed transformation. This is the same indeterminacy as in Shen and Xiu (2026)'s Theorem 3, extended to the time-varying case. In the linear case with* $\varphi_i^\star(F, s) = \lambda_i(s)'F$ *and* $L_N \asymp N$*,* $\epsilon_N \lesssim N^{-1}$ *and the rate simplifies to* $N^{-1}K_1 + T^{-2\beta/(2\beta+K+1)}$*, matching Shen and Xiu (2026) up to the extra time dimension.*

D Proofs

D.1 Proof of Theorem 1

The proof adapts Theorem 2(a) of Shen and Xiu (2026) (henceforth SX). Two changes from SX: the decoder is on $\mathbb{R}^{K_1+1}$ (extra time input), and the encoder approximates $\eta^\star$ on a $(K+1)$-dim box. Several approximation and concentration results are imported from the SX supplementary appendix and stated below.

D.1.1 Notation and function class

Let $\{X_{it}\}$ for $i = 1, \dots, N$ and $t = 1, \dots, T$ be the observed panel, $X_t \in \mathbb{R}^N$ the cross-section, and $s_t = t/T \in (0, 1]$ the scaled time index. Let $\varphi_i^\star : \mathbb{R}^K \times [0, 1] \to \mathbb{R}$ denote the true loading function, $F_t^\star \in \mathbb{R}^K$ the latent factors, U_{it} the error, and $X_{it}^\star = \varphi_i^\star(F_t^\star, s_t)$ the common component.

Definition 1 (Time-conditioned disjoint-output autoencoder class)**.** *For $d_1, d_2, w_1, w_2 \in \mathbb{N}$ and $C, B > 0$, let $\mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, C, B)$ denote the set of pairs $(\rho, \{\psi_i\}_{i=1}^N)$ with $\rho \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, C, B)$ and $\psi_i \in \mathcal{F}_{K_1+1}^{1}(d_2, w_2, C, B)$ for each i, where $\mathcal{F}_{n_0}^{n_{d+1}}(d, w, C, B)$ is the ReLU network class of Shen and Xiu (2026), §2.2. The depth parameter d_1 is an upper bound; the class includes networks of depth $\le d_1$, with shallower networks padded with identity layers.*

The estimator is a minimizer $(\hat\rho, \{\hat\psi_i\}_{i=1}^N)$ of $L(\rho, \{\psi_i\}) := \sum_{t,i}(X_{it} - \psi_i(\rho(X_t, s_t), s_t))^2$ over $\mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, T^{5\beta+5}, B)$. The fitted value is $\hat X_{it}^G := \hat\psi_i(\hat\rho(X_t, s_t), s_t)$. The target quantity is $\mathcal{E}_{NT} := (NT)^{-1}\sum_{t,i}(\hat X_{it}^G - X_{it}^\star)^2$.

D.1.2 Imported results from SX

Proposition 4 (SX App. Proposition A1: Kohler–Langer approximation)**.** *Let $f \in \mathcal{H}^p([-a, a]^r, C)$ for $p = q + s$, $q \in \mathbb{N}_0$, $s \in (0, 1]$, $C > 0$, $a \ge 1$. Let $\zeta \ge \zeta_0(p, a, C, \|f\|_{C^q})$ sufficiently large. If $d \asymp \lceil \log_4(\zeta^{2p})\rceil \cdot (\lceil \log_2(\max\{q, r\}+1)\rceil + 1)$ and $w \asymp 2^r \binom{r+q}{r} r^2 (q+1)\zeta^r$, then there exists a network $\hat f_{\text{wide}}$ with depth d, width w, weights bounded by ζ^{5p+5}, such that $\|\hat f_{\text{wide}} - f\|_\infty \le C'\zeta^{-2p}$ on $[-a/2, a/2]^r$, where C' depends only on $(p, a, C, \|f\|_{C^q})$.*

Lemma 1 (SX App. Lemma 5: sub-Gaussian bound)**.** *Under Assumption 1, for any fixed matrix of constants $\{C_{it}\}$ and conditional on $F^\star$, the statistic $(\sum_{t,i} C_{it}U_{it})/(\sum_{t,i} C_{it}^2)^{1/2}$ is sub-Gaussian with norm bounded by $\sigma_u > 0$, despite cross-sectional and serial dependence.*

Lemma 2 (SX App. Lemma 7)**.** *Under Assumptions 1–3, if $\log T = o(\epsilon_N^{-1})$, then with probability at least $1 - C\exp(-c\epsilon_N^{-1})$, $\|W^\star U_t\|_\infty \le 1$ for all $t \in \{1, \dots, T\}$.*

Lemma 3 (SX App. Lemma 8)**.** *Under Assumptions 1–3, with probability at least $1 - C\exp(-cT)$, $T^{-1}\sum_{t=1}^T \|W^\star U_t\|^2 \lesssim \epsilon_N$.*

Lemma 4 (SX App. Lemma 9)**.** *For $\varphi \in \mathcal{F}_{n_0}^{n_{d+1}}(d, w, T^{5\beta+5})$ and $x, y \in \mathbb{R}^{n_0}$, $|\varphi(x) - \varphi(y)| \le n_0 T^{(5\beta+5)(d+1)} w^d \|x - y\|_\infty$.*

Lemma 5 (SX App. Lemma 10). *For $\mathcal{F}_{n_0}^{n_{d+1}}(d, w, T^{5\beta+5}, B)$ with weight vector $n = (n_0, \ldots, n_{d+1})$ and at most S nonzero weights, there exists finite $\mathcal{F}_{n_0,\delta}^{n_{d+1}}$ with $|\mathcal{F}_{n_0,\delta}^{n_{d+1}}| \le (8\delta^{-1}CT^{(5\beta+5)(d+2)}(1+w)^d n_0 n_{d+1} d)^{2S}$ such that for every φ in the class there exists $\overline{\varphi}$ in the subset with $\|\varphi - \overline{\varphi}\|_\infty \le \delta$ for all $\|x\|_\infty \le C$.*

Lemma 6 (Basic inequality). *For any element $(\phi_i \circ \rho)$ of $\mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, T^{5\beta+5}, B)$, the minimizer $(\hat{\rho}, \{\hat{\psi}_i\})$ of L satisfies*

$$\sum_{t,i} \big(X_{it}^\star - \hat{X}_{it}^G\big)^2 \le \sum_{t,i} \big(X_{it}^\star - \phi_i(\rho(X_t, s_t), s_t)\big)^2 + 2\sum_{t,i} U_{it}\big(\hat{X}_{it}^G - \phi_i(\rho(X_t, s_t), s_t)\big). \quad (13)$$

D.1.3 Auxiliary lemmas

Lemma 7 (Encoder existence). *Suppose Assumptions 1–3 hold. Let $d_1 \asymp \log T$ and $w_1 \asymp T^{(K+1)/(2(2\beta+K+1))}$. Then, with probability at least $1 - C\exp(-cT) - C\exp(-c\epsilon_N^{-1})$, there exists $\rho^\dagger \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, T^{5\beta+5}, B)$ satisfying*

$$\frac{1}{T}\sum_{t=1}^{T} \big\|\rho^\dagger(X_t, s_t) - (F_t^\star, \mathbf{0}_{K_1-K})\big\|^2 \lesssim \epsilon_N + T^{-\frac{2\beta}{2\beta+K+1}}. \quad (14)$$

Proof. Apply the affine rescaling $\tau(z, s) := (z, (B+1)(2s-1))$, mapping $[-2B, 2B]^K \times [0, 1]$ onto $[-2B, 2B]^K \times [-(B+1), B+1]$. Let $g(z, u) := \eta^\star(z, (u/(B+1)+1)/2)$ on that image. Since g is β-Hölder with constant C_η on a box, by the Whitney extension theorem there exists $\tilde{g} \in \mathcal{H}^\beta([-2(B+1), 2(B+1)]^{K+1}, \tilde{C}_\eta)$ with $\tilde{g} = g$ on $[-2B, 2B]^K \times [-(B+1), B+1]$ and $\tilde{C}_\eta$ depending only on (β, K, B, C_η).

Apply Proposition 4 to $\tilde{g}$ with $a = 4B$, $r = K+1$, $p = \beta$, $\zeta = T^{1/(2(2\beta+K+1))}$. There exists $\hat{f}_{\text{wide}}$ with depth $d_1' \asymp \log T$, width $w_1 \asymp \zeta^{K+1} = T^{(K+1)/(2(2\beta+K+1))}$, weights $\le \zeta^{5\beta+5} \le T^{5\beta+5}$, and

$$\sup_{[-(B+1),B+1]^{K+1}} |\hat{f}_{\text{wide}} - \tilde{g}|_\infty \le C_\eta' T^{-\frac{\beta}{2\beta+K+1}}. \quad (15)$$

Applying Proposition 4 K times, one per output coordinate of $\tilde{g}$, and taking the parallel composition, gives a vector-valued $\eta^\dagger$ with total width $Kw_1 \asymp w_1$ and the same approximation bound. Define $\eta^\dagger(z, s) := \hat{f}_{\text{wide}}(z, (B+1)(2s-1))$, clip the output to $[-B, B]^K$ via $C_B(y) := (\min(|y_j|, B)\operatorname{sgn}(y_j))_{j=1}^K$, and set $\rho^\dagger(x, s) := (C_B(\eta^\dagger(W^\star x, s)), \mathbf{0}_{K_1-K})$. Weights are bounded by $T^{5\beta+5}$, so $\rho^\dagger \in \mathcal{F}_{N+1}^{K_1}(d_1, w_1, T^{5\beta+5}, B)$.

Write $z_t := W^\star X_t = W^\star\varphi^\star(F_t^\star, s_t) + W^\star U_t$, and $u_t := (B+1)(2s_t - 1)$. By Lemma 2, $z_t \in [-B-1, B+1]^K$ on a high-probability event, so $(z_t, u_t) \in [-(B+1), B+1]^{K+1}$. Since $F_t^\star \in [-B, B]^K$, $C_B(F_t^\star) = F_t^\star$, and $\|C_B(y) - F_t^\star\| \le \|y - F_t^\star\|$ for any y. Hence

$$\|\rho^\dagger(X_t, s_t) - (F_t^\star, \mathbf{0})\| \le \|\eta^\dagger(z_t, s_t) - F_t^\star\|. \quad (16)$$

Decompose $\eta^\dagger(z_t, s_t) - F_t^\star$ into three terms: $\eta^\dagger(z_t, s_t) - \eta^\star(z_t, s_t)$, $\eta^\star(z_t, s_t) - \eta^\star(W^\star\varphi^\star(F_t^\star, s_t), s_t)$, and $\eta^\star(W^\star\varphi^\star(F_t^\star, s_t), s_t) - F_t^\star$. The first is bounded by $C_\eta' T^{-\beta/(2\beta+K+1)}$. The second, by Lipschitz continuity of $\eta^\star$, is bounded by $KC_\eta\|W^\star U_t\|$. The third is bounded by $2\epsilon_N$ in squared norm. Squaring and applying $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ gives $\|\rho^\dagger(X_t, s_t) - (F_t^\star, \mathbf{0})\|^2 \lesssim T^{-2\beta/(2\beta+K+1)} + \|W^\star U_t\|^2 + \epsilon_N$. Averaging over t and applying Lemma 3 gives the result. □

Lemma 8 (Decoder existence)**.** *Under Assumption 2, there exist* $\psi_i^\dagger \in \mathcal{F}_{K_1+1}^1(d_2, w_2, T^{5\beta+5}, B)$, $i = 1, \dots, N$, *with* $d_2 \asymp \log T$, $w_2 \asymp T^{(K+1)/(2(2\beta+K+1))}$, *and*

$$\sup_{(z,s)\in[-B,B]^{K_1}\times[0,1]} \left|\psi_i^\dagger(z,s) - \varphi_i^\star(z_{1:K}, s)\right| \le CT^{-\frac{\beta}{2\beta+K+1}}, \tag{17}$$

where $\varphi_i^\star$ *is extended from* $\mathcal{B}$ *to* $[-B,B]^{K_1}$ *by applying* $\varphi_i^\star$ *to* $z_{1:K}$.

Proof. Apply $\tau(z,s) := (z, (B+1)(2s-1))$, mapping $[-B,B]^K \times [0,1]$ onto $[-B,B]^K \times [-(B+1), B+1]$. Define $g_i(z,u) := \varphi_i^\star(z, (u/(B+1)+1)/2)$ on that image; $g_i \in \mathcal{H}^\beta([-B,B]^K \times [-(B+1), B+1], 2^\beta B)$. Whitney-extend g_i to $[-2(B+1), 2(B+1)]^{K+1}$, obtaining $\tilde{g}_i \in \mathcal{H}^\beta([-2(B+1), 2(B+1)]^{K+1}, \tilde{C}_{\varphi^\star})$.

Apply Proposition 4 with $a = 2(B+1)$, $r = K+1$, $p = \beta$, $\zeta = T^{1/(2(2\beta+K+1))}$. There exists $\hat{f}_{i,\text{wide}}$ with depth $d_2' \asymp \log T$, width $w_2 \asymp \zeta^{K+1} = T^{(K+1)/(2(2\beta+K+1))}$, weights $\le T^{5\beta+5}$, and sup error $\tilde{C}'_{\varphi^\star} T^{-\beta/(2\beta+K+1)}$. Apply C_B outside, and define $\psi_i^\dagger(z,s) := (C_B \circ \hat{f}_{i,\text{wide}})(z_{1:K}, (B+1)(2s-1))$, zero-padded to input dimension $K_1 + 1$. □

Lemma 9 (Empirical process)**.** *Let* $(\rho, \{\psi_i\}) \in \mathcal{F}_{AE}^{K_1}(d_1, d_2, w_1, w_2, T^{5\beta+5}, B)$ *be arbitrary, and set* $\tilde{X}_{it} := \psi_i(\rho(X_t, s_t), s_t)$, $s_*^2 := \sum_{t,i}(X_{it}^\star - \tilde{X}_{it})^2$. *Under Assumption 1 and the class definition, with probability at least* $1 - C\exp(-cT)$, *simultaneously for all members,*

$$\left|\sum_{t,i} U_{it}(X_{it}^\star - \tilde{X}_{it})\right| \lesssim \left(NT^{\frac{K+1}{2\beta+K+1}} + TK_1\right)^{1/2} \log^{3/2} T \cdot \left(s_* + N^{1/2}\right) + N. \tag{18}$$

Proof sketch. Let $\mathcal{S} \subset \mathbb{R}^{K_1}$ be a grid of spacing $\varepsilon_T := (K_1+1)^{-1} T^{-1-(5\beta+5)(d_2+1)} w_2^{-d_2}$ covering $[-B,B]^{K_1}$. Then $\log|\mathcal{S}| \lesssim K_1 \log^2 T$. Apply Lemma 5 to $\mathcal{F}_{K_1+1}^1(d_2, w_2, T^{5\beta+5}, B)$ at $\delta = T^{-1}$. Nonzero-weight count $S \asymp w_2^2 d_2 \asymp T^{(K+1)/(2\beta+K+1)} \log T$, and log-cardinality $\log|\mathcal{F}_{2,\delta}| \lesssim T^{(K+1)/(2\beta+K+1)} \log^3 T$.

For each tuple $(\overline{\psi}_{\ell_i}, (\overline{F}_{k_1}, \dots, \overline{F}_{k_T})) \in \mathcal{F}_{2,\delta}^N \times \mathcal{S}^T$, the coefficients $C_{it} := X_{it}^\star - \overline{\psi}_{\ell_i}(\overline{F}_{k_t}, s_t)$ are deterministic conditional on $F^\star$. By Lemma 1, $\sum_{t,i} C_{it}U_{it}/\|C\|$ is sub-Gaussian with norm σ_u. The union bound over $|\mathcal{S}|^T \cdot |\mathcal{F}_{2,\delta}|^N$ tuples and the choice $u = 2\sigma_u(T\log|\mathcal{S}| + N\log|\mathcal{F}_{2,\delta}|)^{1/2}$ give a high-probability event on which $|\sum C_{it}U_{it}|/\|C\| \le u$ simultaneously.

The trained $(\rho, \{\psi_i\})$ corresponds to one tuple via nearest-grid-point assignment, with pointwise error $\le 2T^{-1}$. Hence $|\sum U_{it}(X_{it}^\star - \tilde{X}_{it})| \le |\sum U_{it}(X_{it}^\star - \overline{\psi}_{\ell_i}(\overline{F}_{k_t}, s_t))| + 2T^{-1}\sum|U_{it}|$. First term: $u\|\overline{C}\|$ with $\|\overline{C}\| \le s_* + 2N^{1/2}$. Second term: $2T^{-1}\sum|U_{it}| \lesssim N$ with probability $\ge 1 - e^{-cNT}$. □

D.1.4 Assembly

Let $L := T^{-2\beta/(2\beta+K+1)} + N^{-1}K_1 + \epsilon_N$ and $\mathcal{E}_{NT} := (NT)^{-1}\sum_{t,i}(\hat{X}_{it}^G - X_{it}^\star)^2$.

Step 1 (approximation). Let $(\rho^\dagger, \{\psi_i^\dagger\})$ be from Lemmas 7–8. For each (t,i), define $T_1 := \varphi_i^\star(F_t^\star, s_t) - \varphi_i^\star(\rho^\dagger(X_t,s_t)_{1:K}, s_t)$ and $T_2 := \varphi_i^\star(\rho^\dagger(X_t,s_t)_{1:K}, s_t) - \psi_i^\dagger(\rho^\dagger(X_t,s_t), s_t)$. Then $|T_1| \le \sqrt{K}B\|F_t^\star - \rho^\dagger(X_t,s_t)\|$ and $|T_2| \le CT^{-\beta/(2\beta+K+1)}$. Consequently, $(NT)^{-1}\sum_{t,i}(X_{it}^\star - \psi_i^\dagger(\rho^\dagger(X_t,s_t), s_t))^2 \le C_0 L$.

Step 2 (assembly). Apply Lemma 6 with $(\phi_i \circ \rho) = (\psi_i^\dagger \circ \rho^\dagger) \in \mathcal{F}_{AE}^{K_1}$. Divide by NT, split the cross term into two pieces, and apply Lemma 9 to each. With $A_1 := (T^{-2\beta/(2\beta+K+1)} +$

$N^{-1}K_1)^{1/2}\log^{3/2}T$ and $B_1 := (T^{-2\beta/(2\beta+K+1)-1} + N^{-1}K_1T^{-1})^{1/2}\log^{3/2}T + T^{-1}$, there exist constants $c_1, c_2 > 0$ such that $\mathcal{E}_{NT} \le C_0L + c_1A_1\mathcal{E}_{NT}^{1/2} + c_1A_1L^{1/2} + c_2B_1$.

Step 3 (magnitudes). $A_1^2 \le L\log^3 T$. Since $2\beta/(2\beta+K+1) < 1$, $T^{-2\beta/(2\beta+K+1)} > T^{-1}$, so $L \ge T^{-1}$; hence $(L/T)^{1/2} \le L$, giving $B_1 \lesssim L\log^{3/2}T$. Also $A_1L^{1/2} \lesssim L\log^{3/2}T$.

Step 4 (closing). Set $v = \mathcal{E}_{NT}^{1/2}$, $c = c_1A_1$. From $v^2 \le C_0L + 2cv + c_1A_1L^{1/2} + c_2B_1$, completing the square gives $v \le 2c + (C_0L + c_1A_1L^{1/2} + c_2B_1)^{1/2}$, hence $\mathcal{E}_{NT} \le 8c^2 + 2C_0L + 2c_1A_1L^{1/2} + 2c_2B_1 \lesssim L\log^3 T$.

D.2 Proof of Proposition 1

The proof applies Theorem 1 at each grid point with the kernel weight as the effective sample. The local fit at x_j uses $\asymp Th$ observations, and the loading function $\varphi_i^\star(F, s)$ evaluated at $s = x_j$ is still $(K+1)$-dimensional. The nonparametric error at each grid point is $(Th)^{-2\beta/(2\beta+K+1)}$, and the smoothing bias in the s dimension is $h^{2\beta}$. Balancing the two gives $(Th)^{-2\beta/(2\beta+K+1)} \asymp h^{2\beta}$, which yields $h \asymp T^{-1/(2\beta+K+2)}$ and the rate $T^{-2\beta/(2\beta+K+2)}$. The remaining terms are $N^{-1}K_1$ and ϵ_N. Summing over the grid and applying the kernel weighting introduces a log factor, giving the stated $\log^4 T$.

D.3 Proof of Proposition 2

The proof is Theorem 1 with the decoder dimension changed from $K_1 + 1$ to $K_1 + P + 1$, and the loading-function dimension from $K + 1$ to $K + P + 1$. The encoder is unchanged: it still outputs K_1 from (X_t, s_t). Three changes:

Decoder existence. Apply Proposition 4 with $r = K + P + 1$ and $\zeta = T^{1/(2(2\beta+K+P+1))}$. Width $w_2 \asymp T^{(K+P+1)/(2(2\beta+K+P+1))}$; sup error $CT^{-\beta/(2\beta+K+P+1)}$.

Empirical process. The decoder class now has input dimension K_1+P+1 and width $w_2 \asymp T^{(K+P+1)/(2(2\beta+K+P+1))}$, so the entropy bound $\log|\mathcal{F}_{2,\delta}|$ picks up $T^{(K+P+1)/(2\beta+K+P+1)}\log^3 T$. The factor grid $\mathcal{S}$ is unchanged.

Assembly. Steps 2–4 of Theorem 1 proceed verbatim, with $L := T^{-2\beta/(2\beta+K+P+1)} + N^{-1}K_1 + \epsilon_N$.

D.4 Proof sketch of Proposition 3

The argument adapts Shen and Xiu (2026)'s proof of Theorem 3 with $K \to K + 1$ and the encoder output $\hat\rho(X_t, s_t) \in \mathbb{R}^{K_1}$ in place of $\hat\rho(X_t)$.

Reduction to the linear case. The analogue of SX App. Lemma 14 is needed with the encoder evaluated at (X_t, s_t) and the target at $F_t^\star$. This lemma bounds the difference between the in-sample loss and the out-of-sample loss using Bernstein's inequality for the networks in $\mathcal{F}_{N+1}^{K_1}$ and the grid $\mathcal{S} \subset \mathbb{R}^{K_1}$. The bound picks up entropy terms $\log|\mathcal{F}_{N+1}^{K_1}|$ and $\log|\mathcal{S}|$, giving a rate of $(T^{-2\beta/(2\beta+K+1)} + T^{-1/2}L_N^{1/2})\log^3 T$.

Substitution of the encoder. With μ_t defined as the map that sends the encoder output to the factor via the inverse of $W^\star\varphi^\star(\cdot, s_t)$ (extended to a map on $\mathbb{R}^{K_1}$ by zero padding), the

triangle inequality yields

$$\frac{1}{T}\sum_{t=1}^{T}\|\mu_t(\hat{\rho}(X_t,s_t))-F_t^\star\|^2 \lesssim \epsilon_N K_1 + \frac{1}{T}\sum_{t=1}^{T}\|\hat{\rho}(X_t,s_t)-(F_t^\star,\mathbf{0})\|^2\cdot(1+\|\eta^\star\|_\infty^2).$$

The first term is the encoder approximation error from Assumption 3 times the bottleneck dimension. The second term picks up (i) the in-sample encoder rate $T^{-2\beta/(2\beta+K+1)}+\epsilon_N$ multiplied by $N\epsilon_N$, and (ii) a second-order term $N\epsilon_N^2$ from the quadratic approximation of $\eta^\star$ around the noiseless factor. Combining gives the stated rate.

E Minimum-variance portfolio application

This appendix documents the minimum-variance portfolio (MVP) application. The theory bounds the accuracy of the recovered common component, not the accuracy of the precision matrix. The MVP application tests a downstream task the theory does not directly address; the result is negative, and informative about the boundary of what the estimator delivers.

E.1 Setup and methods

The MVP weights depend only on the covariance matrix: $w = \hat{\Sigma}^{-1}\mathbf{1}/(\mathbf{1}'\hat{\Sigma}^{-1}\mathbf{1})$, and the MVP Sharpe ratio depends on how well $\hat{\Sigma}$ approximates the true covariance. The data are the S&P 100 daily panel (497 stocks, 5,029 trading days, 2005–2024), with cleaning that drops rows with missing return or market cap, converts Refinitiv percent returns to decimal, and drops rows with $|r_{it}| \geq 50\%$. Estimation uses a rolling window of $T = 250$ days with monthly rebalancing ($H = 21$ days). The out-of-sample period is 2010–2024.

Nine estimators are compared: equal-weight, sample covariance, Ledoit–Wolf linear shrinkage, Ledoit–Wolf nonlinear shrinkage (OAS), POET, FWYZ (the time-varying local PCA covariance estimator of **fan2024time**), the static autoencoder with POET residual covariance, TV-AE-G with POET residual covariance, and TV-AE-C with $K_1 \in \{3, 5\}$.

E.2 Results

The best performer is Ledoit–Wolf linear shrinkage (1.005), followed by POET (0.984). The autoencoder-based covariances deliver Sharpe ratios of 0.71–0.76, below the linear shrinkage benchmark and roughly comparable to the equal-weighted portfolio. The autoencoder recovers the factor structure (the volatility is in line with PCA and POET), but the recovered structure does not translate into a better MVP Sharpe.

E.3 Why MVP underperforms

The MVP Sharpe is dominated by the smallest eigenvalues of the inverse covariance matrix. The autoencoder's nonlinear correction perturbs these eigenvalues in-sample, and the perturbation does not generalize out of sample. PCA-based covariances preserve the eigenvalue structure MVP optimization is sensitive to. This is a mismatch between the estimand of

Table 4: Out-of-sample Sharpe ratios for minimum-variance portfolios on the S&P 100 panel.

Estimator	Sharpe	Ann. vol.
Equal weight (1/N)	0.838	0.173
Sample covariance	0.855	0.142
Ledoit–Wolf linear	**1.005**	0.128
Ledoit–Wolf nonlinear (OAS)	0.977	0.131
POET	0.984	**0.124**
Static AE + POET	0.761	0.121
TV-AE-G + POET	0.733	0.121
TV-AE-C ($K_1 = 5$) + POET	0.733	0.132
TV-AE-C ($K_1 = 3$) + POET	0.708	0.131
S&P 500	0.730	0.172

the reconstruction problem and the estimand of the portfolio problem, not a failure of the autoencoder. The negative result maps the boundary of when the autoencoder helps, and motivates using a metric (reconstruction R^2) that matches the theorem.

F What did not work

This appendix documents applications that were attempted and not pursued to a successful conclusion, and one implementation issue. The failures are reported because they are informative about what the method can and cannot do at the sample sizes considered.

F.1 Equity return prediction and encoder collapse

The first attempted application was monthly equity return prediction, following Gu, Kelly, and Xiu (2021). Monthly returns were pulled for 3,521 US common stocks (share codes 10 and 11, prices above \$1, market caps above \$100 million), 2000–2024, giving 617,063 observations after dropping rows with missing return or market cap. The first round used a 500-stock slice (top 500 permnos by mean market cap, 2010–2024).

The first specification was TV-AE-G with no intercept term in the decoder, matching the simulation setup. The result was a total out-of-sample R^2 of 0.029 and a predictive R^2 of 0.0029. For reference, PCA on the same panel achieves 0.38 total R^2 and 0.0007 predictive R^2. The autoencoder was an order of magnitude worse than PCA on reconstruction.

The diagnosis was encoder collapse to a constant. The correlation between predicted and fitted returns was 0.999998. Computing the standard deviation of the factor components per test year gave values between 10^{-7} and 10^{-12}: the encoder had collapsed to outputting the per-stock mean return. The signal-to-noise ratio of monthly returns (roughly 10^{-4} per observation) combined with the high dimensionality of the encoder input (1,000 after padding and masking) made gradient descent unable to find the factor structure. Adding an intercept term to the decoder ($r_{it} = \alpha_i + \beta_i(t)'F_t$) did not fix the problem: R^2 remained at 0.016.

The PCA warm-start (Section I) fixes the collapse by initializing the encoder at the PCA loadings. With the warm-start, the autoencoder recovers the factor structure and produces an out-of-sample R^2 in the same range as PCA. The warm-start was not enough to beat PCA on return prediction, for the reason documented next.

F.2 The characteristic ceiling

The next fix was to add characteristics to the model, following Gu, Kelly, and Xiu (2021). Without OSAP data, six price-based characteristics were constructed from the daily returns panel: log market cap, 12-month momentum, 1-month reversal, 12-month volatility, 12-month beta, and 12-month idiosyncratic volatility. Lagged by one day to avoid look-ahead.

The result was still below PCA. In a single estimation window, the fraction of return variance spanned by the six characteristics was 0.0192, and the fraction captured by rank-3 PCA was 0.5463 (0.5908 for rank-5). The characteristics span less than 2% of return variance; any factor model conditioned on these characteristics can capture at most 1.92% of the variance through the characteristic channel. PCA does not suffer this limitation because its factors are constructed to maximize explained variance, not to depend on characteristics. The ceiling is a property of the data and of the six characteristics, not of the estimator. To match Gu, Kelly, and Xiu (2021), the 94-characteristic OSAP panel (or a comparable WRDS construction) is needed.

F.3 Local estimators at $T = 250$

The local autoencoder and local PCA both fail to train at the estimation window used for the reconstruction application ($T_{\text{est}} = 250$). The effective sample per grid point is $Th \approx 37$ observations with the Silverman bandwidth $h \approx 0.148$. Local PCA on the equity volatility panel produced $R^2 = -9.60$, and on the crypto return panel gave $R^2 = 0.026$ even with an inflated bandwidth. The local-estimator failure has a clean theoretical interpretation: the rate $T^{-2\beta/(2\beta+K+2)}$ is asymptotically valid but requires $Th \to \infty$ with $h \to 0$, which at $T = 250$ forces $Th \lesssim 37$. The global estimator avoids this by pooling information across the full T observations.

F.4 The weight-decay asymmetry

An earlier run of the MVP application had the autoencoder-based covariance appearing to outperform PCA-based covariances by a substantial margin on a subset of the out-of-sample period. The cause was that the autoencoder's optimizer had been given a weight-decay parameter while the PCA benchmark had not. The comparison was between a regularized model and an unregularized one, and the regularization was producing a spurious advantage. The lesson is that autoencoder comparisons against linear benchmarks must be symmetric in regularization. This is the same lesson as the simulation-side placebo test, which detected a similar asymmetry in Section 5.1.

G One-sided test for nonlinearity

The comparison across panels can be framed as a one-sided test for the presence of nonlinearity in the loading function. Under the null that the loading function is linear, the autoencoder is over-parameterized, so its expected out-of-sample R^2 is weakly lower than PCA's. Under the alternative, the autoencoder can capture the nonlinear structure, so its expected R^2 can be higher. The test is: reject linearity if the bootstrap CI on $\Delta R^2 = R^2_{\text{AE}} - R^2_{\text{PCA}}$ is entirely above zero; fail to reject otherwise.

The test is asymmetric: rejection is evidence of nonlinearity, non-rejection is not evidence of linearity. Non-rejection can arise because the loading function is linear, because the autoencoder is underpowered at the given sample size, or because the nonlinearity is too small to detect.

On the four panels: non-rejection on equity returns, equity volatility, and crypto volatility; rejection on crypto returns ($\Delta R^2 = +0.036$, 95% CI $[+0.008, +0.075]$). The crypto return panel is the only one where linearity can be rejected at the 5% level.

The test has a known limitation: the alternative includes both time-varying linear loadings and nonlinear loadings, and at this sample size the two channels cannot be distinguished. If the true loading function is $\lambda_i(t/T)'F_t$ (time-varying but linear) and the time variation is strong enough, TV-AE-G can reject linearity by capturing the time variation rather than the nonlinearity. The local PCA analogue would distinguish the channels, but the local estimators do not train at $T = 250$.

H Additional simulation details

H.1 The full DGP grid

The full factorial design crosses five loading-function classes (linear, exponential, quadratic, radial, sigmoid) with five time-variation schemes (constant, single abrupt break, multiple abrupt breaks, smooth logistic, smooth double logistic) and three noise structures (iid, heteroskedastic, cross-sectionally dependent), giving $5 \times 5 \times 3 = 75$ DGPs. The main text reports five. The excluded combinations were run on a subset and produced qualitatively identical conclusions.

The loading functions are: linear $\varphi_i^\star(F) = C\lambda_i' F$; exponential $\varphi_i^\star(F) = C\exp(\lambda_i' F)$; quadratic $\varphi_i^\star(F) = C_1\lambda_{1i}' F + C_2 F'\lambda_{2i} F$; radial $\varphi_i^\star(F) = C\exp(-\|\lambda_i - F\|^2)$; sigmoid $\varphi_i^\star(F) = C/(1+\exp(-\lambda_i' F))$. Scaling constants C, C_1, C_2 are chosen so that the unconditional variance of $X_{it}^\star$ is unit; in the quadratic case, the linear and quadratic terms contribute equally.

H.2 First-pass baseline

The first simulation pass compared PCA, local PCA, and the static autoencoder on seven DGPs at $T = 200$.

Three patterns emerge: local PCA dominates on linear DGPs (correctly specified); the static autoencoder dominates on exponential and quadratic DGPs (captures nonlinearity); neither linear method handles the nonlinear cases. The gap motivates the time-varying autoencoder.

Table 5: First-pass baseline: PCA, local PCA, static AE on seven DGPs ($T = 200$).

DGP	PCA	Local PCA	Static AE
linear, constant	0.035	0.067	0.049
linear, logistic	0.193	0.069	0.086
linear, single_break	0.194	0.093	0.077
linear, multiple_logistic	0.183	0.070	0.079
exponential, constant	0.201	0.504	0.084
exponential, logistic	0.214	0.183	0.144
quadratic, logistic	0.478	0.478	0.170

H.3 Capacity sweep

The sweep varied weight decay $\lambda \in \{0, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$, decoder width $\in \{16, 32, 64\}$, encoder width $\in \{32, 64, 128\}$, and bottleneck $K_1 \in \{2, 3, 4\}$, on the exponential/logistic and quadratic/logistic DGPs. Weight decay shows monotone improvement up to 10^{-3} and degradation above; decoder and encoder widths are flat; $K_1 = 3$ dominates $K_1 = 2$ and is comparable to $K_1 = 4$. The final config uses $K_1 = 3$, encoder 64, decoder 32, weight decay 10^{-3}.

Weight decay is applied symmetrically across all autoencoders. An earlier run applied it asymmetrically and produced a spurious 25% "win" for the time-conditioned autoencoder on the linear/constant placebo DGP. The placebo test is what detected this: since the DGP has no time variation, any advantage had to be an artifact. Under the corrected specification, the two autoencoders tie on the placebo DGP.

I Additional empirical details

I.1 Data construction

Equity panel. Daily returns from Refinitiv, 2005–2024. Cleaning: drop missing return or market cap; convert percent to decimal; drop $|r_{it}| \geq 50\%$ (data errors in daily equity data). Final: 497 stocks, 5,029 trading days. Coverage averages 0.62 across the (i, t) grid.

Crypto panel. 50 coins from Refinitiv, 2015–2024. The composite crypto feed ({`COIN`}`=`) resolves for BTC, ETH, XRP, LTC, BCH. Three suffix conventions were tested: {`COIN`}`=` (composite), {`COIN`}`=CCCL` (CoinDesk), and {`COIN`}`=BTSP` (Bitstamp). Fifty of 58 candidate coins resolved under at least one convention. The remaining 8 (EOS, KSM, ZIL, WAVES, ANKR, STORJ, CELO, EURT) did not resolve and were dropped. Coverage builds over time: the panel reaches 25 coins by mid-2021 and 50 coins by December 2021.

Volatility panels. 5-day realized volatility: $\log \mathrm{RV}_{it} = \frac{1}{2} \log(\sum_{s=0}^{4} r_{i,t-s}^2 + \epsilon)$, with $\epsilon = 10^{-10}$ floor.

Unbalanced panel treatment. Each permno (or coin) is assigned a fixed slot in the encoder input of dimension $N_{\max} = 100$ (equity) or $N_{\max} = 50$ (crypto). Missing values are filled with zero, and a 0/1 mask vector is concatenated with the returns and the time index, giving an encoder input of dimension $2N_{\max} + 1$.

PCA warm-start. The encoder is parameterized as $F_t = W_{\text{PCA}} X_t + \text{MLP}(X_t)$, where $W_{\text{PCA}} \in \mathbb{R}^{K_1 \times N}$ is initialized to the top-K_1 PCA loadings of the training window and the MLP output layer is initialized to zero. Weight decay on W_{PCA} is set to zero. Without the warm-start, gradient descent on monthly or daily returns collapses to the per-stock mean (see Appendix F).

I.2 Choice of K_1

$K_1 = 5$ matches the PCA benchmark and the factor-model literature for equity panels. The theory predicts a linear $N^{-1}K_1$ cost for over-specification, so modest over-specification is harmless. Consistent with the simulations, where $K_1 = 3$ dominates $K_1 = 2$ and $K_1 = 4$ adds nothing. Sensitivity on the crypto panel: the AE's advantage over PCA peaks at $K_1 = 2$ (a gain of 3.4pp) and falls to zero at $K_1 = 12$. If the AE were only overfitting noise, the gain would increase with K_1; instead, it peaks at a finite bottleneck width and declines, consistent with genuine low-dimensional nonlinear structure.

I.3 What the R^2 measures

The pooled out-of-sample R^2 is $1 - \sum_{(i,t)\in\text{OOS}} (\hat{X}_{it} - X_{it})^2 / \sum_{(i,t)\in\text{OOS}} X_{it}^2$. It measures against observed X, not against $X^\star$. Since $X^\star$ is unobservable in real data, this is the natural construction. If $X = X^\star + U$ and $\hat{X}$ recovers $X^\star$ up to error η, then $\|\hat{X} - X\|^2 = \|\hat{X} - X^\star\|^2 + \|U\|^2 - 2\langle \eta, U \rangle$. The cross term is dominated by $\|U\|^2$, so R^2_{OOS} is a monotone transform of the theoretical quantity.

The limitation is that the metric cannot distinguish "captures nonlinear common structure" from "captures cross-sectionally correlated idiosyncratic structure." Both improve R^2 against X. The four-panel comparison and the placebo checks mitigate this by requiring the AE to win only on the panel with suspected nonlinearity.

Shen and Xiu (2026) use MSE against the true $X^\star$ in simulations. Out-of-sample R^2 against X is the analog when $X^\star$ is unobservable; the same construction is standard in the autoencoder literature.